

Ushtrime ND1ç

Të llogaritet shprehja për shpejtësinë fillestare dhe të shprehet në $\frac{dm}{s}$ dhe $\frac{cm}{s}$:

$$v_0^2 = \left(77 \frac{km}{h}\right)^2 - 2 \cdot 55 \cdot 10^{-2} \frac{m}{s^2} \cdot 0.324 km$$

$$\underline{77 \frac{km}{h} = 77 \frac{1000m}{3600s} = \frac{77000m}{3600s} = 21,38 \frac{m}{s}}$$

$$0,324 km = 0,324 \cdot 1000m = \underline{324m}$$

$$v_0^2 = \left(21,38 \frac{m}{s}\right)^2 - 110 \cdot 10^{-2} \frac{m}{s^2} \cdot 324m$$

$$v_0^2 = 457,1044 \frac{m^2}{s^2} - 356,4 \frac{m^2}{s^2} = 100,7044 \frac{m^2}{s^2}$$

$$v_0^2 = 100,7044 \frac{m^2}{s^2} \quad \sqrt{\quad}$$

$$v_0 = \sqrt{100,7044} \cdot \sqrt{\frac{m^2}{s^2}} = \underline{10,03 \frac{m}{s}}$$

$$\boxed{v_0 = 10,03 \frac{m}{s}}$$

$$\boxed{1m = 1 \cdot 10^3 dm}$$

$$1m = 1 \cdot 10^2 cm$$

$$v_0 = 10,03 \cdot \frac{10^3 dm}{s} = 100,3 \frac{dm}{s}$$

$$v_0 = 10,03 \cdot 100 \frac{cm}{s} = \underline{1003 \frac{cm}{s}}$$

Të gjendet $t = ?$ nga shprehja e me poshtme, koha të shprehet në minuta dhe ore?

$$-8520 mm = \underbrace{(0 m/s) \cdot t + 0.5 \cdot (-9.8 m/s^2) \cdot t^2}_{=0}$$

$$\cancel{8,52m} = \cancel{4,9 \frac{m}{s^2} \cdot t^2} \quad / (-1)$$

$$\boxed{1mm = 1 \cdot 10^{-3} m}$$

$$8,52 = 4,9 \frac{1}{s^2} \cdot t^2$$

$$t^2 = \frac{8,52}{4,9} \cdot s^2 = 1,74 s^2 \quad \sqrt{\quad}$$

$$t = \sqrt{1,74 s^2} = \underline{1,32 s}$$

$$\boxed{t = 1,32 s}$$

$$\underline{0,022 min}$$

$$1min = 60s \Rightarrow 1s = \frac{1min}{60}$$

$$t = 1,32 \cdot \frac{1min}{60} = 0,022 min = \underline{2,2 \cdot 10^{-2} min}$$

$$1h = 3600s \Rightarrow 1s = \frac{1h}{3600}$$

$$t = 1,32 \frac{1h}{3600} = 0,00036h = \underline{3,6 \cdot 10^{-4} h}$$